

ZECPATH AI SYSTEM ARCHITECTURE

Day 2 Internship Deliverables

Submitted By

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Internship: Data Science Engineer Intern

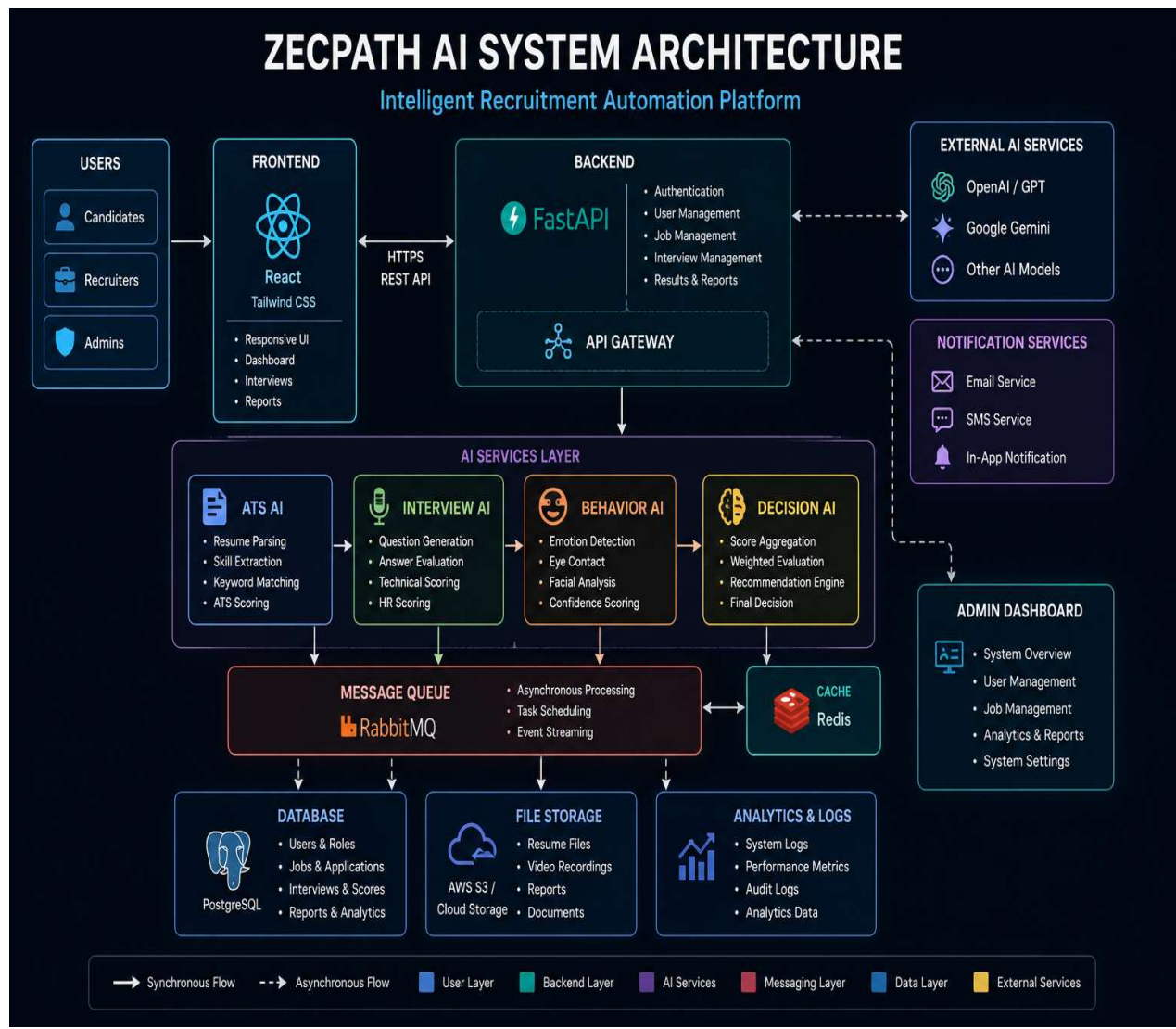
Date: 25 July 2026

Submitted To

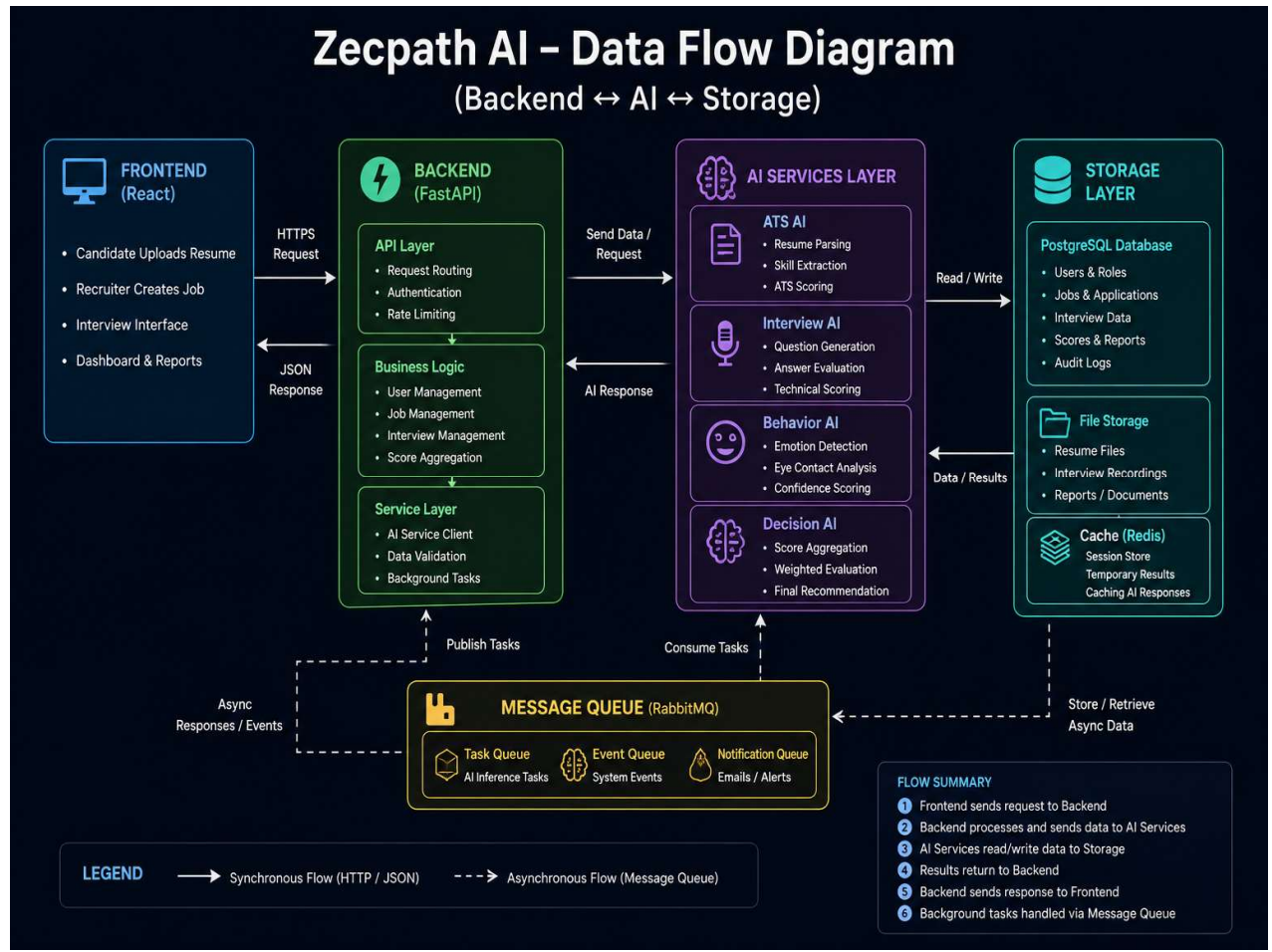
Company: Zecpath

Department: Product & Engineering Team

1. AI system architecture diagram



2. Data flow diagram (Backend ↔ AI ↔ Storage)



3. Input/output specification for each AI engine

<u>AI Engine</u>	<u>Input</u>	<u>Processing</u>	<u>Output</u>
Resume Parser AI	Resume (PDF/DOCX), Candidate Details	Extracts personal information, education, skills, certifications, projects, and work experience using NLP	Structured candidate profile (JSON), extracted skills, experience summary
ATS Scoring AI	Parsed resume, Job Description	Matches resume against job requirements, calculates relevance score, identifies missing skills	ATS score, skill match percentage, strengths, missing skills, shortlist recommendation
Voice Screening AI	Audio response, Interview questions	Speech-to-text conversion, communication analysis, pronunciation, fluency, confidence assessment	Transcript, communication score, fluency score, confidence score
HR Interview AI	Candidate profile, HR responses	Evaluates communication, behavioral answers, leadership, teamwork, problem-solving, and cultural fit	HR evaluation report, competency scores, overall HR rating

Technical Interview AI	Candidate profile, Technical answers, Coding submissions	Generates technical questions, evaluates answers, checks coding logic, accuracy, and technical knowledge	Technical score, coding assessment, strengths, weaknesses, recommendation
Behavioral Analysis AI	Video stream, Audio stream	Detects facial expressions, eye contact, emotions, engagement, and behavioral patterns	Behavior report, emotion analysis, confidence score, engagement score
Machine Test AI	Test responses, Code files, Quiz answers	Executes test cases, checks correctness, performance, and completion	Test score, execution results, performance metrics
Decision AI	ATS score, HR score, Technical score, Behavioral score, Machine test score	Aggregates evaluation results using weighted scoring and decision rules	Final recommendation (Hire, Hold, Reject), overall score, decision summary
Offer Generation AI	Selected candidate details, Job information, Salary details	Generates personalized offer letter and onboarding documents	Offer letter (PDF), onboarding package, notification status
Notification AI	Recruitment events, Candidate status	Sends emails, SMS, or in-app notifications based on workflow events	Delivery confirmation, notification logs, candidate updates